

Talking to the Kernel through Sysfs

The Linux kernel provides a virtual file system called *sysfs*. By providing virtual files, *sysfs* is able to export information about various kernel sub-systems, hardware devices and associated device drivers from the kernel's device model to user space. To further explore *sysfs*, dive deep into this article.



There is a need to provide information related to each process, to the user space, which can then be used by programs such as *ps*. The */proc* file system was created for this purpose. Through the *proc* file system, each process has its directory in the */proc* folder. It was originally designed to provide process related information to user space. Adding directories and files to the */proc* file system is easier; so, many kernel sub-systems started using this file system for displaying information to the user space. It is also used to take inputs from the user space to control settings inside the kernel modules. But the */proc* file system is getting cluttered with lots of non-process related information.

From the Linux 2.5 development cycle, a new interface called the */sys* file system has been introduced. *Sysfs* is a

RAM based file system. It is designed to export the kernel data structures and their attributes from the kernel to the user space, which then avoids cluttering the */proc* file system.

The advantages of *sysfs* over *procfs* are as follows:

- A cleaner, well-documented programming interface
- Automatic clean-up of directories and files, when the device is removed from the system
- The enforced *one item per file* rule, which makes for a cleaner user interface

The 'one item per file' rule mandates that in each file of *sysfs*, there will be only one value that can be put in or read from it. This feature really makes it a cleaner interface. Through *sysfs*, user space programs can get information from the kernel sub-system like device drivers. Programs